

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☒	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☒ A description of all covariates tested
☒	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☒	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Data collection were performed using Python (version 3.8.19).
Data analysis	The models were implemented using Python (version 3.8.19) with the following libraries: NumPy (version 1.23.5), Pandas (version 2.0.3), Betacal (version 1.1.0), Matplotlib (version 3.7.5), and Scikit-learn (version 1.3.0). Statistical analyses were conducted using R (version 4.4.0) (R Foundation for Statistical Computing, Vienna, Austria). Categorical variables were compared using the χ^2 test or Fisher exact test, as appropriate, while continuous variables were analyzed using the Student's t-test or one-way analysis of variance (ANOVA), as appropriate. Statistical significance was set at a two-tailed P value of < 0.05.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The data are not publicly available due to restrictions from KTDB policy and belong to the National Emergency Medical Center (NEMC) of Korea. NEMC provides de-

identified KTDB to researchers for nonprofit academic research. Details of accessing the raw data and guide are available at the NEMC website (<https://dw.nemc.or.kr>). The validation dataset used in this study contains potentially sensitive patient information and, therefore, will be made available upon reasonable request to ensure compliance with ethical guidelines and institutional privacy policies.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	The study cohort comprised participants of both sexes, as detailed in the demographic table. Sex was self-reported by participants, while data on gender identity were not collected. Analyses evaluating the effect of sex on study outcomes were conducted and are presented in Supplementary Materials.
Reporting on race, ethnicity, or other socially relevant groupings	The AI model was developed using a dataset derived from a Korean population. Information on race, ethnicity, or other socially relevant groupings was not available in either the development dataset or the external validation datasets from Korean and Australian trauma centers. Consequently, these variables were not included in the analysis.
Population characteristics	Population characteristics are summarized in Table 1. This study included trauma patients from the Korean Trauma Data Bank (KTDB), consisting of a training set (n=163,351) and a test set (n=40,838), as well as external validation datasets from four Korean trauma centers (n=8,358) and one Australian trauma center (n=3,578). Due to IRB restrictions, patients under 18 years of age were excluded from the Australian dataset. Emergency department (ED) mortality was 0.9% in both KTDB sets, 1.0% in the Korean external validation set, and 0.1% in the Australian dataset (p<0.001). In-hospital mortality ranged from 3.5% to 4.4% across datasets (p=0.085). The mean Injury Severity Score (ISS) was 10.5 ± 9.2 in KTDB, 10.5 ± 9.1 in the Korean external set, and 8.7 ± 7.8 in the Australian dataset (p<0.001). The mean age was 56.7 ± 22.4 years in the Korean external validation set and 51.4 ± 22.4 years in the Australian dataset (p<0.001). Male patients comprised 65.2% of the KTDB test set, 64.7% of the training set, 61.2% of the Korean external validation set, and 66.4% of the Australian dataset (p<0.001). Regarding injury intent, accidental injuries accounted for 92.5% in the KTDB test set, 87.1% in the Korean external dataset, and 68.3% in the Australian cohort. The proportions of self-harm, assault, and unspecified intent varied (p<0.001). Falls were reported in 18.7% of KTDB test set cases and 46.4% of Australian cases. Slips accounted for 24.9% of KTDB cases but were not reported in the Australian dataset. Motor vehicle-related injuries, including car and motorcycle accidents, were present in all datasets, with varied proportions across regions. Injury mechanism categories showed significant differences (p<0.001 for most comparisons). Most injuries were classified as blunt trauma (e.g., 92.6% in KTDB test set, 91.0% in Australia), with penetrating injuries and burns reported less frequently (p<0.001). Accident locations included homes, general roads, sports facilities, and commercial areas. The proportion of unknown locations was 32.7% in the Australian dataset and 21.8% in the Korean external set (p<0.001). Job-related injury status was not recorded in the Australian cohort (100% missing). Hospital arrival route and mode, including direct visit, ambulance use, and air transport, were comprehensively recorded in Korean datasets but not reported in the Australian dataset (100% missing). Prehospital vital signs such as systolic and diastolic blood pressure, pulse rate, respiratory rate, body temperature, and SpO₂ were documented in the KTDB and Korean external sets but mostly missing in the Australian dataset. For instance, SBP measurements were missing in 75.1% of Australian cases (p<0.001). The AVPU scale showed that 40.0% of KTDB test set patients and 12.7% of Australian patients were "alert" on arrival. Missing AVPU data accounted for 74.5% in the Australian cohort (p<0.001). Protective equipment use was reported in Korean datasets but entirely unrecorded in the Australian set. Among KTDB patients, 15.9%–16.3% reported use of protective gear, with 51.0% marked as unknown (p<0.001).
Recruitment	The AI model was developed using data from the Korean Trauma Data Bank (KTDB), a national trauma registry managed by the Korean National Emergency Medical Center. The KTDB includes trauma data from 19 centers across South Korea, comprising all 17 regional trauma centers, which serve as the highest-level trauma centers, and 2 trauma training centers. The dataset used for model training spanned the years 2017 to 2022 and included 204,189 patients. For external validation, the model was further tested on independent datasets obtained in 2023 from four regional trauma centers in South Korea and from 2022 to 2024 at one Level 1 trauma center in Australia. The South Korean validation sites included Chonnam National University Hospital (CNUH, n=1,675), Cheju Halla General Hospital (CHGH, n=2,309), Chungbuk National University Hospital (CNUH, n=1,277), and Gachon University Gil Medical Center (GUGC, n=3,097). The Australian validation dataset was obtained from Westmead Hospital (WH, n=3,578). A total of 11,936 trauma patients were included in the external validation dataset.
Ethics oversight	This study was approved by the IRBs of five participating institutions. Ethical approval was obtained from CHGH (CHH-2024-LO3), CNUH (CNUH 2024-10-017), GUGC (GUGC2024-357), CNUH (CNUH-2024-016), and WH (2024/PID0326). The requirement for informed consent was waived due to the retrospective nature of the study.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

- ☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	This study utilized a large-scale, multicenter cohort from the Korean Trauma Data Bank (KTDB), a national trauma registry maintained by the Korean National Emergency Medical Center, comprising a total of 204,189 patients. For external validation, data were collected from four South Korean trauma centers: Chonnam National University Hospital (CNUH, n=1,675), Cheju Halla General Hospital (CHGH, n=2,309), Chungbuk National University Hospital (CBUH, n=1,277), and Gachon University Gil Medical Center (GUGC, n=3,097). Additionally, an external validation dataset was obtained from Westmead Hospital (WH) in Australia (n=3,578).
Data exclusions	From the Korean Trauma Data Bank (KTDB), a total of 227,567 patients were initially identified. Patients were excluded if they were transferred to other facilities (n=19,340), had non-trauma-related deaths (n=23), were discharged with medical futility (n=22), refused cardiopulmonary resuscitation (CPR) (n=3,991), or had missing data on age (n=2). After applying these criteria, 204,189 patients were included for model development and internal validation. For external validation, data were collected from four regional trauma centers in South Korea and one level 1 trauma center in Australia, comprising a total of 12,635 patients. Exclusion criteria included inter-facility transfer (n=423), refusal of CPR (n=161), missing data on symptom onset time (n=0), and missing age information (n=7). In addition, 108 patients under the age of 18 were excluded from the Australian dataset in accordance with IRB restrictions. The final external validation cohort consisted of 11,936 patients.
Replication	To ensure reproducibility of the findings, the model was developed using a large-scale nationwide trauma registry (Korean Trauma Data Bank, KTDB) following a prespecified analytical protocol. Internal validation was conducted using a temporally independent subset of the KTDB data. External validation was performed using independent datasets from four regional trauma centers in South Korea and one trauma center in Australia. The model's performance was consistently evaluated across these cohorts using predefined performance metrics. All attempts at replication were successful, and no findings were identified that could not be reproduced.
Randomization	Not applicable. This study was observational in nature, and participants were included based on convenience sampling without random assignment.
Blinding	Blinding was not applicable, as this was a retrospective observational study without any intervention.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	Not applicable. This study did not involve plants or seed stocks.
Novel plant genotypes	Not applicable. No novel plant genotypes were generated or analyzed in this study.
Authentication	Not applicable.